

A scenic mountain landscape with green hills, rocky cliffs, and a cloudy sky. The foreground shows a rocky, grassy slope. In the middle ground, there are steep, rocky cliffs and green hills. The background features more distant, hazy mountain ranges under a sky filled with white and grey clouds.

AI & Software Engineering Workshop

Week 1, Day 1: Setup and First Message

Edik Simonian, Summer 2026

Hi, I'm Edik 🙌

- **Lead Software Engineer at NASA JPL 🚀** — 9 years
- **B.S. from UC Davis · M.S. in Computer Science from Georgia Tech**
- My **second year** teaching here at **TUMO Yerevan**



First, let's meet the room

Go around, **30 seconds each**:

- **Who are you?** Your name and grade
- **What computer science classes have you taken?** Any programming course — in school, a club, or online — or none yet
- **What are you hoping to learn from this class?** A skill, a kind of project, or just whatever drew you in

No wrong answers, even "nothing yet" is fine. This is where we find out what to build by Friday.

Ground rules

- **No YouTube, gaming, or phone calls** during the workshop
- **Quick quizzes** at the start of each workshop, and again after the break
- **Don't be more than 5 minutes late** — a quiz you miss is one you can't take
- **No copy-pasting code from ChatGPT** — use the Claude Code we provide, and understand every line you ship

See it working first

t.me/tele_pythonanywhere_bot

This is where you'll be by **Friday**: your own bot, your personality, live on the internet.

Go ahead, message it now, then we'll build one from scratch.

What you'll build this week

- A **Telegram bot** powered by a real LLM
- With its own **personality**, you design it
- Custom **slash commands**, you code them
- **Memory** that survives restarts
- **Live on the internet** by Friday, running even when your computer is off

By the end of today: the bot replies to you, from code on your machine.

The week, day by day

- **Day 1, Setup & First Message** (*today*): tokens, `make run`, your first `/start` edit
- **Day 2, Personality & Prompts**: a system prompt that gives your bot a voice — and meet Claude Code
- **Day 3, New Commands**: live-code `/joke`, then build your own, with a test
- **Day 4, Memory & Deploy**: memory that survives restarts, then go live on PythonAnywhere
- **Day 5, Build Your Own Bot**: design & ship a feature that's all yours, then demo

Each day builds on the last — by Friday it's yours, and online.

What is a bot?

Two ways Telegram talks to your code:

- **Polling**: your code keeps asking Telegram "*any new messages?*"
→ what we use **today**, on your computer
- **Webhook**: Telegram pushes each message to your server's URL
→ what we use **Day 4**, in production

Same bot code either way. Only the delivery changes.

The stack

Telegram → Flask (Python) → Cerebras → reply

- **Telegram Bot API**: the messaging interface
- **Flask**: receives the messages
- **Cerebras**: runs the LLM that writes the replies (*free tier*)
- **SQLite**: memory (*Day 4*)
- **PythonAnywhere**: hosting (*Day 4*)
- **GitHub**: your code, your tests, your deploys

First: open your terminal

*This lab runs **Windows**, so we work inside **WSL** — real Linux, running on Windows. Every command this week then runs exactly as the slides show.*

1. Open the **Ubuntu** app from the Start menu → a black terminal opens
2. First launch may ask for a **username + password** — pick simple ones, and remember the password
3. **Type every command in this Ubuntu window** — not PowerShell, not Command Prompt

On your own Mac or Linux laptop? You're already set — skip to Setup 1.

Get WSL ready

Work in your **home** folder, never the Windows `C:` drive:

```
cd ~  
pwd          # should say /home/...  – not /mnt/c/...
```

Each line should print a version — if one's missing, tell us:

```
git --version  
python3 --version  
make --version  
gh --version  
curl --version | head -1
```

Stuck? `setup/WSL.md` has every fix — browser won't open, `\r` errors, clock drift after sleep.

Setup 1 of 3: GitHub

1. Create a GitHub account *(if you don't have one)*
2. **Fork** [EdikSimonian/telegram-pythonanywhere-bot](https://github.com/EdikSimonian/telegram-pythonanywhere-bot) — your own copy, needed for auto-deploy on Day 4
3. Clone your fork and install:

```
git clone https://github.com/<your-username>/telegram-pythonanywhere-bot.git
cd telegram-pythonanywhere-bot
make install
```

Setup 2 of 3: Telegram bot token

1. In Telegram, search for **@BotFather**
2. Send `/newbot`
3. Pick a name, then a username ending in `bot`
4. Copy the **token** (looks like `7123456789:AAF...`)

Setup 3 of 3: AI API key

1. Sign up at **cloud.cerebras.ai** (*free, no credit card*)
2. Profile icon → **API Keys** → **Create new API key**
3. Copy the key (looks like `csk-...`)

Wire it up: `.env`

```
cp .env.example .env
```

Set two lines:

```
TELEGRAM_BOT_TOKEN=<your BotFather token>  
AI_API_KEY=<your Cerebras key>
```

Leave the rest as-is.

Secrets live in `.env`. Never in code, never in git.

First contact

```
make run
```

```
Bot @your_bot_username starting in polling mode.  
Send your bot a message on Telegram to try it out.
```

Message your bot. Watch the exchange appear in your terminal.

"Stateless mode" is expected. Memory arrives on Day 4.

Prove it works, like an engineer

```
make test
```

- The whole suite runs **offline**: no API keys, no network
- The **same suite** runs in GitHub Actions on every push to your fork
- Green check on GitHub = your bot's logic still works

Your first code edit

bot/handlers.py:

```
@bot.message_handler(commands=["start"], func=is_allowed)
def cmd_start(message):
    bot.send_message(
        message.chat.id,
        "Hello! I'm your AI assistant...",
    )
```

Change the greeting → Ctrl+C → make run → send /start.

You just shipped your first change.

Today → Tomorrow

Today the bot works on your computer and answers as a generic assistant.

Tomorrow: the most powerful single line in the project, the **system prompt**. Your bot gets a personality.